

SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID - SIH26165
- Problem Statement Title - AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports
- Theme - Smart Automation
- PS Category - Software
- Team ID - _____
- Team Name - Signal-0



Working prototype: <https://sanket-frontend.onrender.com>

SANKET — From Incident Narratives to Early Safety Warnings

SANKET analyzes safety / near-miss reports using AI to detect potential Serious Injury & Fatality (SIF) precursors.

Identifies

- Hazard & Life-Saving Rule
- Control status
- Severity (1–5)
- SIF Precursor: Yes / No
- Evidence & reasoning

Innovation

- LLM + ML hybrid approach
- Explainable risk assessment
- One-Change severity rule
- Prioritized safety-review queue

HOW A REPORT IS JUDGED

Gate	Question	Possible answers
1 Hazard	Is a high-energy hazard present?	yes · no · insufficient information
2 Control	Was a safety barrier doing its job?	absent · failed · present · unclear
3 Severity	How bad could it realistically have been?	1 to 5

Tech Stack

Layer	Technology	Purpose
Frontend	React	worker + admin screens
Backend	FastAPI (Python)	REST API, validation
AI model	Groq GPT-OSS-20B	reads the report
Fallback	TF-IDF + LogReg	answers if AI is down
Database	Supabase (PostgreSQL)	reports + predictions
Cache	SQLite	on-instance, repeat calls

Flow

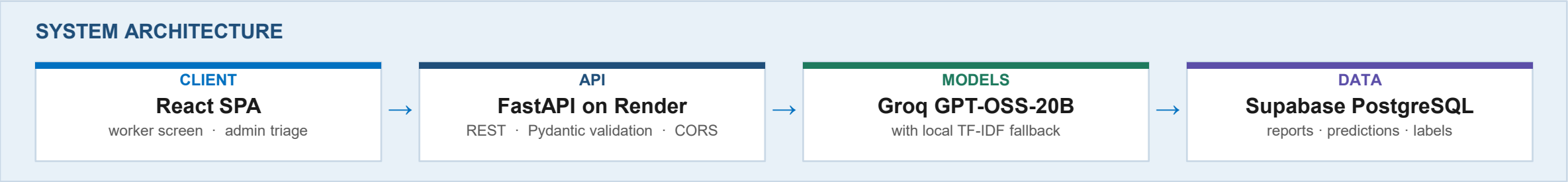
Report
↓
AI / NLP
↓
Hazard
↓
LSR
↓
Control
↓
Severity
↓
SIF Decision
↓
Risk Priority

SIF Rule

Hazard = YES
+ Control = Absent / Failed
+ Severity ≥ 4

→ SIF

All three, or it is not flagged. The flag is computed from the gates, never typed.



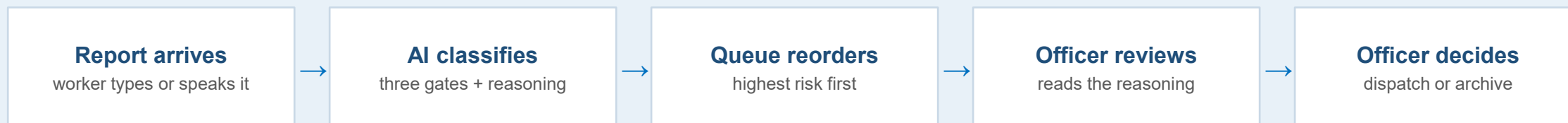
Feasible because

- Uses existing incident reports
- No specialized hardware
- API-based & scalable
- LLM + local fallback
- Human-in-the-loop

CHALLENGES → SOLUTIONS

Challenge	Solution
Unstructured text	LLM reads it against a written rubric
Ambiguous controls	4-state classification, silence ≠ absent
API failure or rate limit	Retry inside a deadline, then local model
Domain variation	Tested against 30 real OSHA narratives, not only our own
Model version changes	Dashboard scopes to one version at a time

HUMAN-IN-THE-LOOP — the system ranks, a person decides



SANKET never closes a report. It only changes the order they are read in.

Potential impact on target audience

- Early detection of SIF precursors in OIL safety reports
- Prioritized review of high-risk incidents
- Faster identification of recurring hazards across sites / activities

Benefits

- Safety: proactive intervention before serious incidents
- Operational: reduces manual screening effort
- Economic: prevents costly injuries and downtime
- Scalable: handles large volumes of reports

MEASURED ON OUR PROTOTYPE DATA

What	Value	Meaning
Labelled corpus	180 reports	150 written for the prototype + 30 real OSHA narratives
Human reviews	360	every report read by two people against one written rubric
Agreed gold labels	173	both reviewers reached the same verdict on all three gates
Reviewer agreement	97% (kappa 0.947)	on a 38-report sample reviewed independently

Safety Reports → SIF Precursors → Risk Prioritization → Early Action → Safer Operations

Prototype evaluation

Measure	Result	How it was measured
Human agreement ceiling	97% · kappa 0.947	two reviewers, working separately, n=38
TF-IDF baseline (F1)	0.754 ± 0.077	5-fold cross-validation, n=114

The ceiling is the bar: two people applying the same rubric to the same reports agreed 97% of the time.

References

Source	What we take from it
DEKRA — Martin & Black (2015)	the SIF precursor concept (cited by the problem statement)
Edison Electric Institute	SIF Precursor model (cited by the problem statement)
VelocityEHS (2024)	PSIF classifier (cited by the problem statement)
IOGP Report 459	Life-Saving Rules — our eight hazard categories
OSHA Severe Injury Reports	30 real narratives, used to check we generalise beyond our own data

Live prototype: <https://sanket-frontend.onrender.com>

Code: github.com/sukanya-2006/SANKET